

5. Aryabhat's Enigma (20 marks)

Problem designed by Apoorva Bhagwat and Angikar Ghoshal

Lenny was studying number theory from an old Sanskrit tome, when he came across the following curious shloka (left, transcribed using Latin script from the original Devanagari script):

<i>tukā nātha kabīra ete ādikavayah </i>	<i>Tuka, Nath and Kabir were the original poets.</i>
<i>nāmā gadimā kintu dwijakavīdvayah </i>	<i>Nama and Gadima were both born twice.</i>
<i>gula ravi thākūra gālība trimūrtī </i>	<i>Gulzar, Ravi Thakur and Galib were a trio.</i>
<i>bahiṇā ekamevā caturastrā kavayitrī </i>	<i>Bahina was the sole versatile poet.</i>

Lenny soon realized that the shloka was referring to several Indian poets: Tukaram, Eknath, Kabir, Namdeo, G. D. Madgulkar (*gadimā*), Gulzar, Ravindranath Thakur, Mirza Galib and Bahinabai Chaudhari. Being a good student of Sanskrit, Lenny promptly translated the shloka, and came up with this slightly nonsensical translation (shown in the right above).

Realizing that there was something fishy about this shloka, Lenny took it to Prof Arya Bhat, and asked him what the shloka meant. “Ah, a classic Aryabhata numeration shloka”, said the professor, pulling out a piece of paper from his drawer. “Each poet represents an integer with a different number of distinct prime factors”, he said.

Poet	Integer	Number of distinct prime factors	Poet	Integer	Number of distinct prime factors
<i>tukā</i>	160001	1	<i>gula</i>	30050	3
<i>nātha</i>	37	1	<i>ravi</i>	6040	3
<i>kabīra</i>	2341	1	<i>thākūra</i>	10052	3
<i>nāmā</i>	45	2	<i>gālība</i>	5026	3
<i>gadimā</i>	1828	2	<i>bahiṇā</i>	10038	4

“The numbers are represented by the words according to a scheme called the *Aryabhata numeration*. I do not exactly remember how the scheme works, but I’m confident the numbers on this paper are correct. Perhaps you can try to reverse-engineer the scheme!” While you do not need to know the Devanagari script to decipher the scheme (the Latin transcriptions have all the information you need), the organization of the letters in the Devanagari alphabet might be useful:

Vowels: *a ā ī ī u ū ṛ ṛ ! ! e ai o au*

Consonants: *k kh g gh ṇ c ch j jh ñ ṭ ṭh ḍ ḍh ṇ t th d dh n p ph b bh m y r l v ś ṣ s h*

Assignment 1 [Answer-Booklet]: Write down the numbers corresponding to the following words:

a) *jāveda*, b) *kālī*, c) *tulasī*, d) *dāsa*, e) *sāhira*

Assignment 2 [Answer-Booklet]: Write down words representing the following numbers (if there are multiple valid representations, any single one is acceptable)

f) 230019, g) 7070, h) 2100017

Assignment 3: Explain the Aryabhata numeration scheme.